

Autoimmune diseases can be caused by a combination of genetic susceptibility and environmental factors. Please see the linked articles for details and complete references.

[Article: Celiac](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-DQ2.5	rs2187668	CC	T	0.11	HLA-DQ2.5 type
HLA-DQ8	rs7454108	TT	C	0.1	HLA-DQ8 type, celiac is possible
HLA- DQ	rs4713586	AA	A	0.94	AA only: DQ2.2 - with below 2 alleles
HLA- DQ	rs2395182	GT	T	0.78	DQ2.2 with other alleles
HLA- DQ	rs7775228	TC	C	0.13	DQ2.2 with other alleles
LPP	rs1464510	--	A	0.44	Increased celiac risk (only with HLA type)
intergenic	rs842647	AA	A	0.64	Increased celiac risk (only with HLA type)
LOC105371664	rs2816316	AA	A	0.8	Increased celiac risk (only with HLA type)
intergenic	rs917997	TC	T	0.24	Increased celiac risk (only with HLA type)
intergenic	rs6441961	TC	T	0.3	Increased celiac risk (only with HLA type)
SH2B3	rs3184504	CC	T	0.45	Increased celiac risk (only with HLA type)
TAGAP	rs1738074	--	T	0.43	Increased celiac risk (only with HLA type)
FUT2	rs601338	--	A	0.47	Increased risk of celiac (only with HLA type)

[Article: HLA-B27](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA	rs4349859	GG	A	0.02	Higher likelihood of carrying HLA-B27
Inter-gen	rs13202464	--	G	0.07	Higher likelihood of carrying HLA-B27

[Article: CTLA4](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CTLA4	rs231775	GA	G	0.37	Increased risk of autoimmune diseases
CTLA4	rs3087243	GG	G	0.57	Increased risk of autoimmune diseases

[Article: Psoriasis](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-C	rs10484554	CT	T	0.14	Increased risk of psoriasis
HLA-C	rs1265181	GC	C	0.2	Increased risk of psoriasis
HLA-B*5701	rs2395029	TT	G	0.03	Increased risk of psoriasis
IL23	rs11209026	GG	A	0.06	Decreased risk of psoriasis
PTPN22	rs1217414	--	A	0.3	AA: increased risk of psoriasis
L12B	rs4085613	TG	T	0.37	Increased risk of psoriasis
IL-17F	rs763780	TT	C	0.06	Increased risk of psoriasis, increased risk of rheumatoid arthritis, increased risk of autoimmune thyroid disease
HLA-C	rs4406273	GA	A	0.08	HLA-Cw*0602; increased relative risk of psoriasis
TNF	rs1799724	CC	T	0.11	TT and CT: anti-TNF-alpha biologic less likely to be effective; CC: anti-TNF-alpha biologic more likely to be effective
TNF	rs1800629	GG	A	0.15	Higher TNF levels, not as good of a response to anti-TNF biologics
TNF	rs361525	GG	A	0.05	Higher TNF levels, increased risk of psoriasis, not as good of a response to anti-TNF biologics
TNFRSF1B	rs1061622	TT	G	0.23	Increase TNF receptor; not as good of a response to anti-TNF biologics

[Article: Rheumatoid Arthritis](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLADRB1	rs660895	AA	G	0.19	High risk for RA
PTPN22	rs2476601	GG	A	0.08	Increased risk for RA
TRAF1	rs10818488	--	A	0.48	Increased risk for RA (2-fold)
TRAF1	rs2900180	--	T	0.33	Increased risk for RA
STAT4	rs7574865	GG	T	0.22	Increased risk for RA
TNFAIP3	rs6920220	AG	A	0.18	Increased risk for RA (2-fold)
CCR6	rs3093024	AA	A	0.42	Slightly Increased risk for RA
ANG2	rs12674822	--	T	0.42	Increased risk of RA (2-fold)

[Article: POTS](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SLC6A2	rs7194256	CC	T	0.15	Increased risk of POTS
GNB3	rs5443	CC	T	0.33	Increased risk of POTS
NOS3	rs2070744	TT	C	0.34	CC: Decreased Risk of POTS
ADRB2	rs1042713	GA	A	0.38	AA: Increased norepinephrine response

[Article: PTPN22](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PTPN22	rs2476601	GG	A	0.08	Increased risk of autoimmune diseases
PTPN22	rs6679677	CC	A	0.08	Increased susceptibility to autoimmune diseases
PTPN22	rs1310182	--	G	0.48	GG only: Increased risk autoimmune diseases

[Article: FMF](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MEFV	rs61732874	--	A	0.002	Familial Mediterranean fever mutation
MEFV	rs3743930	GG	G	0.03	Familial Mediterranean fever mutation
MEFV	rs104895083	--	C	0.00009	Familial Mediterranean fever mutation
MEFV	rs104895094	--	C	0.006	Familial Mediterranean fever mutation
MEFV	rs28940580	--	G	0.0001	Familial Mediterranean fever mutation
MEFV	rs28940578	--	T	0.00004	Familial Mediterranean fever mutation
MEFV	rs11466023	GG	A	0.01	Familial Mediterranean fever mutation
MEFV	rs104895097	--	T	0.00009	Familial Mediterranean fever mutation
MEFV	rs28940579	--	G	0.002	Familial Mediterranean fever mutation
MEFV	i4000409	--	A	0.002	Familial Mediterranean fever mutation
MEFV	i4000403	--	C	0.00009	Familial Mediterranean fever mutation
MEFV	i4000407	--	C	0.006	Familial Mediterranean fever mutation
MEFV	rs61752717	--	C	0.0003	Familial Mediterranean fever mutation
MEFV	i4000406	--	C	0.0003	Familial Mediterranean fever mutation
MEFV	i4000410	--	T	0.00009	Familial Mediterranean fever mutation

[Article: Lupus](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-DQA1	rs2187668	CC	T	0.11	HLA-DRB1*0301. 2-fold increase in risk for lupus.
TNXB	rs1150754	CC	T	0.12	2-fold increase in lupus risk
TNF	rs1800629	GG	A	0.15	Some increase in Lupus risk
TNFAIP3	rs5029939	CC	G	0.07	2x Increased risk of lupus
STAT4	rs7574865	GG	T	0.22	Increased risk of discoid lupus
STAT4	rs10181656	--	G	0.21	Increased risk of lupus
IRF5	rs3807306	--	T	0.46	Increased risk of lupus
IRF8	rs2280381	--	C	0.36	Decreased risk of lupus
IFIH1	rs1990760	CC	T	0.43	Increased risk of lupus
BLK	rs13277113	AG	A	0.26	B lymphoid tyrosine kinase, increased risk of lupus
BLK	rs2248932	AG	A	0.36	B lymphoid tyrosine kinase, increased risk of lupus
IRF5	rs10954213	--	A	0.59	Increased risk of lupus
TLR7	rs179010	TT	T	0.3	Increased relative risk of lupus
SLC15A4	rs1385374	CC	T	0.1	Increased relative risk of lupus
SLC15A4	rs1385374	CC	T	0.1	Increased relative risk of lupus

[Article: Alopecia Areata](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CTLA4	rs1024161	TC	T	0.41	Increased risk of alopecia areata
CTLA4	rs231775	GA	G	0.37	Increased risk of several autoimmune diseases; increased risk of alopecia areata
PTPN22	rs2476601	GG	A	0.08	3 to 4-fold increased risk of alopecia areata
HLA	rs660895	AA	G	0.19	HLA-DRB1*0401; increased risk of alopecia areata
IL17F	rs763780	TT	C	0.05	CC: increased risk of alopecia areata
IL17RA	rs879577	TC	T	0.24	Decreased risk of alopecia areata
IL18	rs1946518	TT	T	0.4	T/T: lower risk of AA
IL2	rs7682241	TG	T	0.34	increased risk of alopecia areata
IL2RA	rs3118470	CC	T	0.69	increased risk of alopecia areata
MIF	rs755622	--	G	0.77	Decreased risk of alopecia areata
TNF	rs1800629	GG	A	0.15	Higher TNF; increased risk of alopecia areata

Article: Raynauds					
Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
NOS1	rs527590	--	T	0.19	Increased susceptibility to Raynaud's syndrome
IL1B	rs1143634	GG	A	0.22	A/A: lower risk of antisynthetase syndrome (good)
IFNG	rs2069718	--	G	0.57	GG: increased risk of Raynaud's in connective tissue disorders
TNF	rs1800630	CA	A	0.15	Increased risk of Raynaud's in lupus patients
HTR1B	rs6297	--	C	0.11	increased risk of Raynaud's due to hand/arm vibrations
IRX1	rs11748327	CC	T	0.27	increased risk of Raynaud's

[Article: Ehlers-Danlos Syndrome](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
COL5A1	rs863223483	--	G	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223478	--	T	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223475	--	I		Mutation linked to possibility of classical EDS
COL5A1	rs863223466	--	A	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223469	--	I	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223470	--	I	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223471	--	I	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223472	--	D	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223473	--	I	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223474	--	D	0	Mutation linked to possibility of classical EDS
COL5A1	rs794727114	--	C	0	Mutation linked to possibility of classical EDS
COL5A1	rs794727760	--	D	0	Mutation linked to possibility of classical EDS
COL5A1	rs80338764	--	C	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223444	--	A	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223445	--	A	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223448	--	C	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223452	--	A	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223453	--	C	0	Mutation linked to possibility of classical EDS
COL5A1	rs863223454	--	T	0.00005	Mutation linked to possibility of classical EDS
COL5A2	rs863223501	--	T	0	Mutation linked to possibility of classical EDS
COL5A2	rs863223495	--	T	0	Mutation linked to possibility of classical EDS
COL5A2	rs863223491	--	T	0	Mutation linked to possibility of classical EDS
COL3A1	rs397509369	--	A	0	Mutation linked to possibility of vascular EDS
COL3A1	rs553203474	--	A	0	Mutation linked to possibility of vascular EDS
COL3A1	rs397509377	--	I	0	Mutation linked to possibility of vascular EDS
COL3A1	rs587779417	--	A	0	Mutation linked to possibility of vascular EDS
COL3A1	rs587779418	--	A	0	Mutation linked to possibility of vascular EDS
COL3A1	rs587779419	--	A	0	Mutation linked to possibility of vascular EDS
COL3A1	rs587779421	--	A	0	Mutation linked to possibility of vascular EDS
COL3A1	rs587779424	--	A	0	Mutation linked to possibility of vascular EDS
COL3A1	rs587779427	--	T	0	Mutation linked to possibility of vascular EDS
COL3A1	rs587779428	--	T	0	Mutation linked to possibility of vascular EDS
COL3A1	rs587779429	--	C	0	Mutation linked to possibility of vascular EDS
COL3A1	rs587779432	--	A	0	Mutation linked to possibility of vascular EDS
COL3A1	rs587779434	--	A	0	Mutation linked to possibility of vascular EDS
COL5A1	rs13946	TC	T	0.74	decreased risk of tendon or ligament injury, reduced relative risk of carpal tunnel syndrome (good! does not cause EDS)
COL5A1	rs61735045	GG	A	0.04	A/A: rare, may cause EDS
TNXB	rs368512272	--	A	0.00005	Classic-like EDS possible

Article: Inclusion Body Myositis

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-DRB1	rs2187668	CC	T	0.11	HLA-DRB1*03:01 alleles, increased risk of IBM
GNE	rs62541771	--	A	0.0001	Carrier of an inclusion body myopathy 2 mutation
GNE	rs28937594	--	G	0.0001	Carrier of an inclusion body myopathy 2 mutation
VCP	rs121909331	--	T	0	Rare mutation linked to myopathies
VCP	rs121909330	--	A	0	Rare mutation linked to myopathies
VCP	rs121909334	--	T	0.002	Rare mutation linked to myopathies
VCP	rs121909335	--	T	0	Rare mutation linked to myopathies

Article: Genetic risk and Guillain-Barré syndrome

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TLR4	rs4986790	AA	G	0.05	Increased relative risk of GBS, up to 8-fold increase for GG
ICAM1	rs1799969	GG	A	0.09	4-fold increased relative risk of GBS
CD1E	rs1065457	--	A	0.59	A/A: common genotype, higher relative risk of GBS in some studies

Article: Pernicious Anemia

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PTPN22	rs6679677	CC	A	0.08	Increased risk of autoimmune diseases and pernicious anemia
AIRE	rs74203920	CC	T	0.01	Increased risk of pernicious anemia
IL2RA	rs2476491	--	A	0.72	Increased risk of pernicious anemia
HLA-DRB1	rs3135388	GG	A	0.11	tagging SNP for HLA-DRB1*15:01; possibly increased risk of pernicious anemia
TCN2	rs9606756	AA	G	0.11	possibly reduced B12, more likely to have pernicious anemia with autoimmune gastritis
CBLIF	rs150884181	AA	G	0.008	intrinsic factor deficiency possible (rare)

[Article: Sjogren Syndrome](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-DRB1	rs2187668	CC	T	0.11	HLA-DRB1*03:01 allele; increased risk of Sjogren, as well as other autoimmune disorders
HLA-DRA	rs3135394	AA	G	0.1	3-fold increased risk of Sjögren's
HLA-DQB1	rs3129716	TT	C	0.11	3-fold increased risk of Sjögren's
HLA-DQA1	rs9271588	CT	C	0.46	half the normal risk of Sjögren's (protective)
IRF5	rs4728142	GG	A	0.42	slightly increased risk of Sjögren's
IRF5	rs10954213	--	A	0.59	A/A: slightly higher risk of Sjögren's
IL12A	rs485497	--	A	0.49	slightly increased risk of Sjögren's
TNIP1	rs6579837	--	T	0.09	slightly increased risk of Sjögren's
STAT4	rs10168266	CC	T	0.19	increased risk of Sjögren's
STAT4	rs11889341	CC	T	0.22	increased risk of Sjögren's
OAS1	rs10774671	--	G	0.36	decreased risk of Sjögren's (protective)
GTF2I	rs117026326	CC	T	0.01	2 to 3-fold increased risk of anti-SSA positive Sjögren's
TNFAIP3	rs2230926	TT	G	0.05	increased risk of Sjögren's; increased risk of lymphoma in Sjögren's patients, especially in younger ages
TNF	rs1800629	GG	A	0.15	Higher TNF-alpha levels. Increased risk of Sjögren's syndrome

Article: Multiple Sclerosis

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-DRB1	rs3135388	GG	A	0.11	HLA-DRB1*1501; up to 3x - 6x increase in risk for MS
HLA-G	rs4959039	AA	G	0.27	1.5x increased risk of MS
C6orf10	rs3129934	CC	T	0.15	Increased risk of MS
IL2RA	rs12722489	CC	C	0.86	decreased risk of MS; reduced interaction of IL2RA with estrogen receptor alpha
IL2RA	rs2104286	TT	C	0.23	C/C: decreased risk of MS
IL7R	rs6897932	CC	C	0.75	Increased risk of MS
IL7R	rs987107	--	A	0.27	A/A: Increased risk of MS
CYP27B1	rs703842	AG	G	0.32	Decreased risk of MS
TNFRSF1A	rs1800693	CT	C	0.39	Slightly increased risk of MS
CD6	rs17824933	CC	G	0.2	Increased risk of MS
IRF8	rs17445836	AG	A	0.2	Decreased risk of MS
IL12B	rs2546890	AA	A	0.5	Slightly increased risk of MS
CD86	rs9282641	GG	A	0.08	Slightly decreased risk of MS
EVI5	rs11808092	CA	A	0.24	Slightly increased risk of MS
IDO1	rs7820268	CT	C	0.67	C/C: 1.5-fold increased relative risk of MS
ARNTL	rs3789327	--	G	0.5	G/G: increased relative risk of MS
CLOCK	rs6811520	--	C	0.67	C/C: increased relative risk of MS
FADS2	rs174611	--	C	0.26	Decreased relative risk of MS
FADS2	rs174618	TC	C	0.42	Decreased relative risk of MS
CD58	rs2300747	AA	G	0.15	Decreased relative risk of MS

Article: Naltrexone

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
OPRM1	rs1799971	GG	G	0.13	More responsive to naltrexone for reducing drinking, reduced opioid receptors; increased pain
TRPM3	rs6560200	--	C	0.5	C/C: Higher risk of CFS/ME (common), Naltrexone may act to restore impaired TRPM3 ion channel activity
TRPM3	rs1891301	--	T	0.51	T/T: Higher risk of CFS/ME, Naltrexone may act to restore impaired TRPM3 ion channel activity
AKR1C4	rs3829125	--	G	0.13	Reduced function, reduced breakdown of naltrexone

[Article: Vaccines and Autoimmune Diseases](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-DRB1*0401	rs660895	AA	G	0.19	high risk of rheumatoid arthritis (~6-fold); possibly increased risk of autoimmune/inflammatory syndrome induced by adjuvants (ASIA)
HLA-DRB1*15:01	rs3135388	GG	A	0.11	increased risk of MS
IL2	rs2069763	CA	A	0.31	high antibody response to measles vaccination; increased risk of autoimmune hepatitis
PTPN22	rs2476601	GG	A	0.08	increased risk of autoimmune diseases including: vitiligo, Graves' disease, type 1 diabetes, Meniere's disease, juvenile arthritis
IL-17F	rs763780	TT	C	0.05	increased risk of rheumatoid arthritis, increased risk of autoimmune thyroid disease

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